

The Two Faces of Algorithmic Collusion: Memory, Spuriousness, and a Model-Free Entropy Diagnostic in Homogeneous-Good Bertrand Competition

Shubh Lamba*

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Abstract

Whether the supra-competitive prices reached by pricing algorithms reflect genuine tacit collusion or a spurious learning artifact is central to competition policy. We study this in homogeneous-good Bertrand competition—the environment most hostile to collusion—contrasting a stateful ε -greedy Q-learner with a stateless mean-based bandit, and classifying outcomes with a model-free, *convergence-validated* Shannon-entropy diagnostic. Three results emerge. First, the *stateless* learner reaches supra-competitive prices immediately and almost always (90–96% of runs) but *weakly* (collusion index $\Delta \approx 0.13$); the *stateful* Q-learner instead converges to competitive pricing at economically plausible horizons (0–2% collusive at $T = 3 \times 10^4$, despite $\approx 90\%$ of its runs converging) and only gradually learns to collude as the horizon lengthens (25%, 44%, 50% at $T = 10^5, 3 \times 10^5, 10^6$), but its collusion is *intense* (Δ rising to 0.46). Agent *memory*, not sophistication, is thus the operative margin, and it generates a reliability-versus-intensity trade-off. Second, a forced-deviation experiment shows the two coordinations differ in kind: the stateless learner is structurally *unable to punish* (rival response amplitude 0.00), so its supra-competitive pricing is *spurious*, whereas the Q-learner punishes deviations sharply (rival price $5.61 \rightarrow 4.05$ against a no-deviation band of 0.45), so its coordination is *genuine*. Third, because the stateless learner converges, the mean-based advantage is *not* a non-convergence artifact—the standard critique of this literature. The policy reading is detection-centred: a model-free screen must flag supra-competitive pricing regardless of an algorithm’s apparent sophistication, because the simplest learner coordinates most reliably and least detectably.

Keywords: algorithmic collusion; reinforcement learning; agent memory; entropy; emergent coordination; competition policy. **JEL:** C63, C73, D43, D83, L13, L41.

1 Introduction

The delegation of pricing to autonomous algorithms has made one question first-order for competition authorities: when such algorithms charge supra-competitive prices, is this genuine tacit

*Independent researcher. Email: lambashubh95@gmail.com. All code, configuration files, random seeds, and generated output are openly available (see Data and Code Availability). Every figure and statistic is reproducible from fixed seeds by the scripts `roadb_experiments.py` and `simulation.py`.

collusion that an effects-based law should reach, or a spurious learning artifact of no cartel significance? Empirically the stakes are real—Assad et al. [2024] find that algorithmic-pricing adoption in German retail gasoline raised margins by roughly 28%. Theoretically the question is sharp: Calvano et al. [2020] show that Q-learning agents in *differentiated*-good markets learn reward–punishment-supported collusion, while a critical literature argues that much apparent algorithmic collusion is an under-exploration “mistake” that vanishes with more exploration or longer horizons [Abada and Lambin, 2023, Abada et al., 2024, den Boer et al., 2024], and that whether learning escapes the competitive equilibrium turns on what agents *condition on* rather than on raw sophistication [Asker et al., 2024, Possnig, 2023].

We take this debate to the hardest case—*homogeneous*-good Bertrand competition, where the static equilibrium is competitive and the temptation to undercut is undiluted by product differentiation—and make the comparison turn on a single primitive: agent *memory*. We contrast a *stateful* ε -greedy Q-learner, whose state is the rival’s previous price (so it *can* condition on, and punish, deviations), against a *stateless* mean-based bandit, which values each price by its historical average payoff and cannot condition on anything. We classify every run with a model-free Shannon-entropy diagnostic into competitive, collusive, or chaotic regimes, and—crucially—we validate convergence *non-circularly* (by greedy-policy stability, not by low entropy) so that the comparison is not confounded by one learner simply failing to settle.

Our findings (Section 6) overturn the natural “sophistication drives collusion” reading and replace it with a memory-based one.

(1) Reliability versus intensity. The stateless learner reaches supra-competitive prices in 90–96% of runs and does so immediately, but *weakly*: its collusion index is $\Delta \approx 0.13$ (on a $[0, 1]$ scale from the discrete Bertrand–Nash benchmark to joint monopoly). The stateful Q-learner, at economically plausible horizons, instead *converges to competition*—only 2% of runs collude even though 89% converge—and learns to collude only slowly as the horizon grows (0%, 25%, 44%, 50% at $T = 3 \times 10^4$ to 10^6), but when it does, its collusion is *intense* (Δ rising to 0.46). Memory, not sophistication, is the operative margin.

(2) Spurious versus genuine. A frozen-policy forced-deviation experiment shows the two coordinations are different in kind. The stateless learner, being unable to condition on the rival, does not respond to a deviation at all (rival price response amplitude 0.00): its supra-competitive pricing is *structurally spurious*, in the sense of Calvano et al. [2023] and the “spontaneous coupling” of Banchio and Mantegazza [2023]. The Q-learner punishes a deviation sharply—the rival’s price falls from 5.61 to 4.05, against a no-deviation noise band of only 0.45—so its coordination is *genuine*, reward–punishment-supported tacit collusion.

(3) The convergence-artifact critique does not apply. Because the stateless learner *converges* (it is fully policy-stable) and still colludes, its advantage cannot be dismissed as a failure to learn—the standard objection to tabular results [den Boer et al., 2024, Abada et al., 2024]. We further trace the horizon dependence of the Q-learner directly, to 10^6 periods, rather than asserting convergence.

The policy implication is detection-centred. Both spurious and genuine supra-competitive pricing raise prices and harm no one’s ability to charge them; yet the spurious variety is produced by the *simplest* learner, arises *fastest*, and—lacking any punishment dynamics—is the hardest to distinguish from competition by behavioural signatures. A model-free screen that flags supra-competitive *outcomes*, independent of the firm’s algorithm, is therefore the appropriate instrument; our entropy diagnostic is one such screen. The remainder of the paper presents the model (Section 3), the learners (Section 4), the diagnostic (Section 5), the results (Section 6), and a discussion of scope and

limitations (Sections 7–8).

2 Related work

Our study sits within the algorithmic-collusion literature founded by Calvano et al. [2020] and Klein [2021], who establish that Q-learning agents autonomously learn reward–punishment-supported supra-competitive pricing in differentiated-product and sequential Bertrand settings respectively. We depart from this canon in three ways: we study the harder homogeneous-good case on a coarse memory-one grid; we add a model-free, convergence-validated entropy classifier that sorts runs into competitive, collusive, and chaotic regimes; and we contrast a stateful Q-learner against a stateless mean-based bandit. The closest and most demanding neighbours warn that supra-competitive pricing may be an artifact rather than strategy. Abada and Lambin [2023] and Abada et al. [2024] trace it to imperfect exploration and under-convergence (“collusion by mistake”); den Boer et al. [2024] argue Q-learning collusion is fragile and emerges only on economically irrelevant timescales. Rather than contest these critiques, we adopt them as our methodological baseline—conditioning every collusion statistic on validated convergence over horizons up to 10^6 periods—and then draw two distinctions they do not. First, reliability versus intensity: the stateless bandit reaches supra-competitive prices more often but at lower normalized collusion index, while the Q-learner colludes less often but harder. Second, a genuine-versus-spurious test indexed by learner class: a forced-deviation diagnostic shows the bandit’s coordination—consistent with the “spontaneous coupling” of Banchio and Mantegazza [2023] and the spurious category of Calvano et al. [2023]—is structurally non-punishable, whereas converged Q-learning is reward–punishment-supported. Crucially, our mean-based result is *not* refuted by Hartline et al. [2024]: their theorem shows external/mean-based no-regret learning is *insufficient* for competition and can permit supra-competitive prices—it never implies mean-based learners become competitive. Our finding is a concrete, finite-horizon, tabular, ε -greedy instantiation of exactly that permitted regime, and Dolgoplov [2024]’s stateless-defection theorem applies to our bandit, not to our memory-enabled Q-learner. We thereby complement Asker et al. [2024] and Possnig [2023] by isolating learner *memory*, not update information, as the operative margin. Our regime taxonomy is in the spirit of the evolutionary oligopoly dynamics studied in this journal by Bischi and Lamantia [2022], though our coordination is emergent from learning rather than hard-wired into agent types.

3 Model

Two firms $i \in \{1, 2\}$ play a repeated Bertrand game with a homogeneous good. Each period each firm chooses a price from a finite evenly spaced grid

$$\mathcal{P} = \{p_{\min}, p_{\min} + \Delta, \dots, p_{\max}\}, \quad \Delta = \frac{p_{\max} - p_{\min}}{K-1},$$

with $p_{\min} = 0$, $p_{\max} = 10$, and K price points; we use $K = 21$ as the baseline and $K = 7$ as a coarse-grid contrast, and treat K as an object of study. Marginal cost is $c = 1$. Demand is inelastic and normalised to one unit, allocated by the Bertrand rule: the lowest-priced firm serves the unit (ties split), so firm i ’s profit is $\pi_i = (p_i - c) q_i$ with $q_i = \mathbf{1}[p_i = \min_j p_j] / |\arg \min_j p_j|$. Firms interact for T periods (baseline $T = 3 \times 10^4$; varied to 10^6).

The discrete competitive benchmark. On a discrete grid the static Bertrand–Nash outcome has both firms at the lowest grid price weakly above cost, which earns a small but *positive* per-firm profit unless a grid point coincides with c . We therefore anchor the collusion index on the true discrete benchmark $\pi^N = (p^{\text{comp}} - c)/2$ with $p^{\text{comp}} = \min\{p \in \mathcal{P} : p \geq c\}$ (so $\pi^N = 0.33$ at $K = 7$ and $\pi^N = 0$ at $K = 21$, where $p = 1$ lies on the grid), and on joint monopoly $\pi^M = (p_{\text{max}} - c)/2 = 4.5$. The *collusion index* of a run is

$$\Delta = \frac{\bar{\pi} - \pi^N}{\pi^M - \pi^N} \in [0, 1], \quad (1)$$

where $\bar{\pi}$ is mean per-firm profit over the final $T_0 = 2000$ periods; $\Delta = 0$ is competitive and $\Delta = 1$ is symmetric joint monopoly. This corrects a common mis-anchoring (setting $\pi^N = 0$) and we compute Δ only on *converged* runs (Section 5).

4 Learning algorithms

Stateful Q-learning. Firm i keeps a table $Q_i(s, a)$, with state s_t equal to the rival’s *previous* price; it plays ε -greedily and updates

$$Q_i(s_t, a_t) \leftarrow Q_i(s_t, a_t) + \alpha \left[\pi_{i,t} + \gamma \max_{a'} Q_i(s_{t+1}, a') - Q_i(s_t, a_t) \right], \quad (2)$$

with $\alpha = 0.10$, $\gamma = 0.95$, and annealed exploration $\varepsilon_t = \varepsilon_0/(1 + \delta t)$, $\varepsilon_0 = 0.10$, $\delta = 3 \times 10^{-4}$ [Watkins and Dayan, 1992]. Conditioning on s_t is what allows reactive, punishment-based strategies.

Stateless mean-based learning. The benchmark keeps the running average payoff $M_i(p)$ of each price and plays ε -greedily with respect to it—a multi-armed bandit with no state, no discount, and no temporal-difference bootstrap. It cannot condition on the rival, so it cannot punish. To rule out an exploration-schedule confound we run both learners under *identical* schedules (Section 6).

5 A convergence-validated entropy diagnostic

We classify outcomes from action logs alone. Over the final T_0 periods we compute each firm’s K -normalised Shannon entropy $H_i = -\sum_p f_i(p) \log_2 f_i(p) / \log_2 K \in [0, 1]$ with a Miller–Madow bias correction, and average to $\bar{H}$; let $\bar{p}$ be the mean price. A run is *competitive* if $\bar{H} < H^*$ and $\bar{p} < 2c$, *collusive* if $\bar{H} < H^*$ and $\bar{p} \geq 2c$, and *chaotic* if $\bar{H} \geq H^*$. Rather than assert a threshold, we *derive* H^* as the antimode of the pooled $\bar{H}$ distribution, obtaining $H^* = 0.288$.

Non-circular convergence. The standard objection to tabular collusion results is that one learner has not finished learning. We therefore certify convergence *independently of entropy*: a run has converged if each firm’s greedy policy ($\arg \max_a Q_i(s, \cdot)$ over visited states, or $\arg \max_p M_i$) is unchanged over the final 10% of the run. All collusion statistics below are reported both unconditionally and on converged runs.

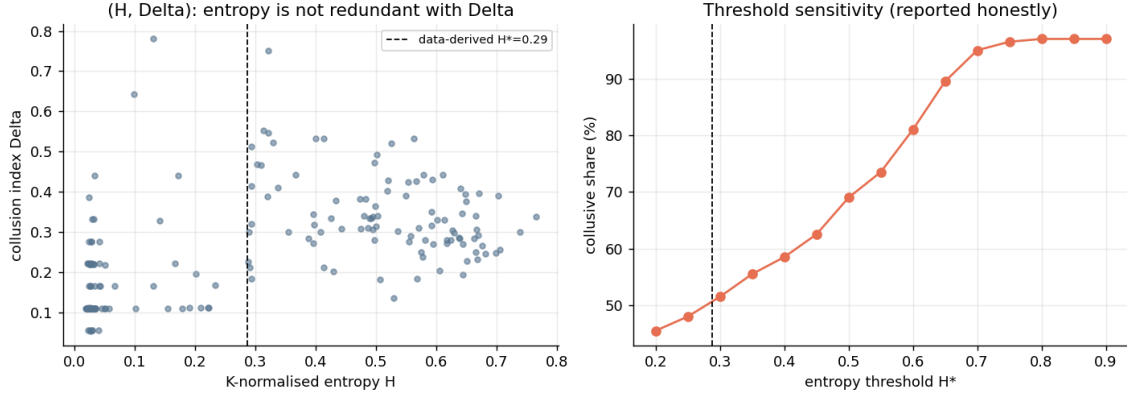


Figure 1: Left: the $(\bar{H}, \Delta)$ plane—entropy is non-redundant with the collusion index, separating converged (low $\bar{H}$) from chaotic play. Right: collusive share vs. the entropy threshold H^* , reported honestly; the dashed line is the data-derived antimode $H^* = 0.288$.

Table 1: Baseline comparison (100 runs/cell). “Conv.” is the policy-stability convergence rate; “Coll.” is collusive share; “Coll. (conv.)” restricts to converged runs; Δ is the collusion index on converged runs.

Grid	Learner	Conv. (%)	Coll. (%)	Coll. (conv.) (%)	Δ (conv.)
$K = 21$	Mean-based (stateless)	91	94	93	0.14
	Q-learning (stateful)	89	2	2	0.35
$K = 7$	Mean-based (stateless)	99	86	86	0.20
	Q-learning (stateful)	48	52	67	0.41

Honesty about thresholds and redundancy. The collusive share is sensitive to H^* (it ranges from 46% at $H^* = 0.20$ to 97% at $H^* = 0.80$ in the pooled sample), so we do not claim threshold robustness; we report the data-derived value and the full sensitivity (Figure 1, right). The diagnostic is *not* redundant with Δ : entropy separates converged from chaotic play, which a price- or profit-based index cannot (Figure 1, left).

6 Results

Unless noted, each cell is 100 independent seeded runs at the baseline ($K = 21$, $T = 3 \times 10^4$).

6.1 Reliability versus intensity, and the death of the artifact critique

Table 1 reports the core comparison. At the baseline grid the stateless learner is collusive in 94% of runs but only weakly ($\Delta = 0.14$), while the Q-learner is collusive in just 2% of runs—*despite* 89% of its runs converging. The Q-learner therefore does not fail to learn; it *learns to compete*. Restricting to converged runs barely moves the figures (93% vs. 2%), so the gap is not a non-convergence artifact (Figure 2). On the coarse $K = 7$ grid the Q-learner colludes far more (52%; 67% among converged) at higher intensity ($\Delta = 0.41$), confirming that grid resolution—how finely the learner can approximate the competitive best response—is itself decisive.

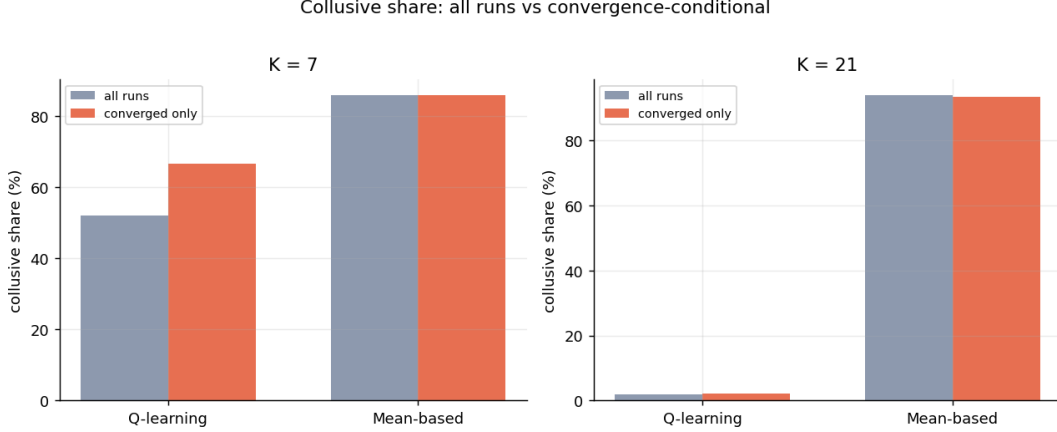


Figure 2: Collusive share, all runs vs. converged-only, at $K = 7$ and $K = 21$. The mean-based advantage survives convergence-conditioning: the Q-learner converges (mostly to competition) rather than failing to learn.

Table 2: Horizon ladder at $K = 21$. Collusive share, convergence rate, and intensity Δ for each learner; seeds per cell 24/24/16/10.

T	Mean-based			Q-learning		
	Coll. (%)	Conv. (%)	Δ	Coll. (%)	Conv. (%)	Δ
3×10^4	96	100	0.13	0	92	0.32
10^5	96	96	0.13	25	75	0.35
3×10^5	94	100	0.13	44	69	0.40
10^6	90	100	0.12	50	70	0.46

6.2 Robustness to the exploration schedule

The contrast is not an artifact of annealing. Under *identical* schedules the mean-based learner dominates throughout: with annealed exploration the collusive shares are 94% (mean) vs. 2% (Q), and with constant $\varepsilon = 0.10$ they are 86% vs. 0%. Holding the schedule fixed isolates memory, not exploration, as the driver.

6.3 The horizon ladder: does the Q-learner catch up?

Because the critical literature stresses that Q-learning collusion may need economically irrelevant horizons, we trace the comparison to $T = 10^6$ at $K = 21$ (Table 2, Figure 3). The mean-based share is essentially flat and high (96% $\rightarrow$ 90%) with stable, weak intensity ($\Delta \approx 0.13$). The Q-learner’s collusive share rises monotonically from 0% to 50%, and its intensity rises with it ($\Delta : 0.32 \rightarrow 0.46$): given enough time the stateful learner increasingly discovers collusion, and the collusion it discovers is *strong*. But even at 10^6 periods the gap does not close, and—as the next subsection shows—the two coordinations remain different in kind throughout.

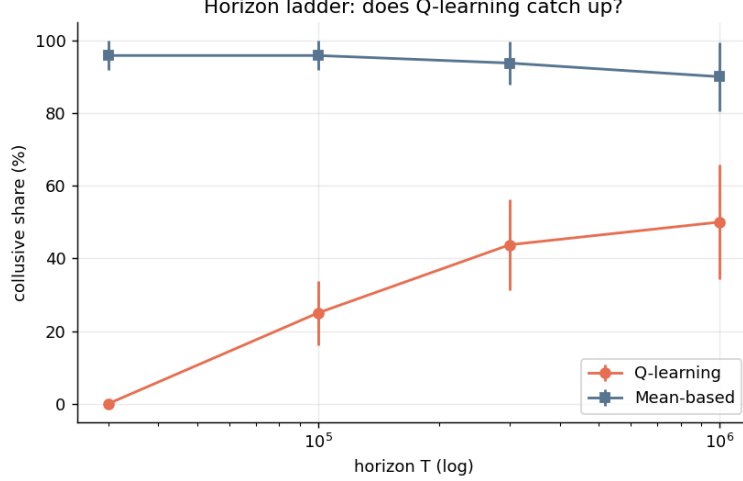


Figure 3: Horizon ladder ($K = 21$, ± 1 s.e.). Mean-based collusion is immediate and flat; Q-learning collusion rises with the horizon but does not overtake it.

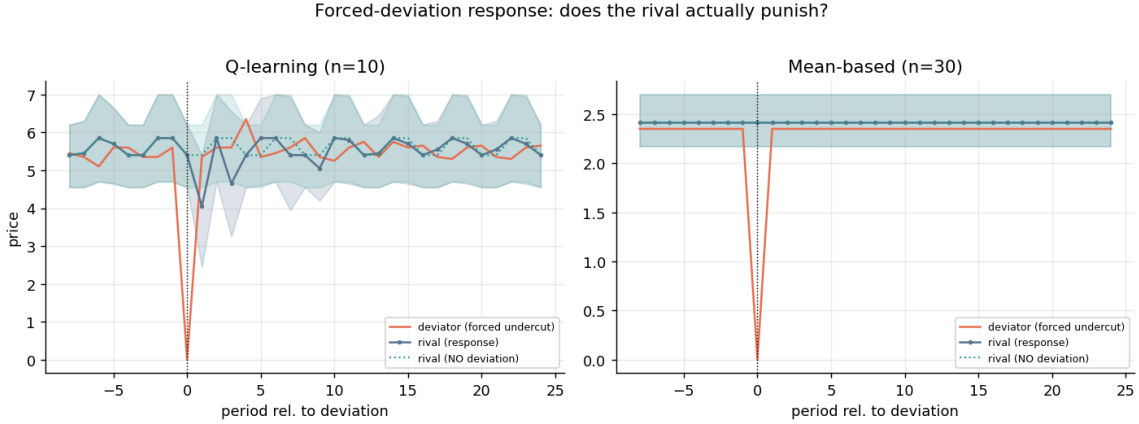


Figure 4: Forced-deviation response (frozen policy). Q-learning (left): the rival punishes a deviation well beyond the no-deviation band. Mean-based (right): no response—spurious coordination.

6.4 Genuine versus spurious: the forced-deviation test

The decisive distinction is mechanistic. We take converged collusive runs, freeze the learned policy, force one firm to undercut for a single period, and trace the rival's response, comparing it to a no-deviation counterfactual with bootstrap bands (Figure 4). For the Q-learner the rival punishes sharply: its price falls from a pre-deviation 5.61 to a trough of 4.05—a drop of 1.56, more than three times the no-deviation band amplitude of 0.45—before cooperation recovers. For the mean-based learner the rival's response amplitude is exactly 0.00: being stateless, it cannot condition on the deviation and does not react. The stateless learner's supra-competitive pricing is therefore *structurally spurious*; the Q-learner's is *genuine*, reward–punishment-supported collusion.

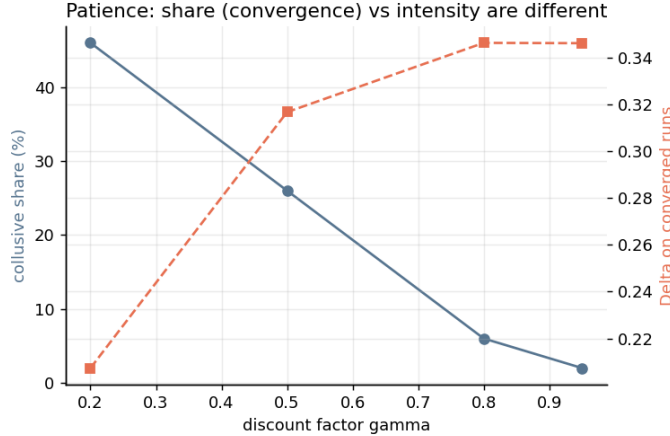


Figure 5: Discount factor at $K = 21$: collusive *share* (left axis) and *intensity* Δ (right axis) move in opposite directions.

6.5 Patience raises intensity, not reliability

Finally, the role of the discount factor differs from the folk-theorem intuition once memory and grid are handled correctly. At $K = 21$, increasing γ from 0.20 to 0.95 *lowers* the Q-learner’s collusive share (46% $\rightarrow$ 26% $\rightarrow$ 6% $\rightarrow$ 2%) while *raising* its intensity on the runs that do collude (Δ : 0.21 $\rightarrow$ 0.32 $\rightarrow$ 0.35 $\rightarrow$ 0.35): patient learners more reliably find the competitive equilibrium, but when they do collude they sustain it harder (Figure 5). We report share and intensity separately rather than conflating them into a single “patience drives collusion” claim.

7 Discussion

Three implications follow. First, in the homogeneous-good limit the relevant primitive is agent *memory*, not sophistication: the learner that *cannot* react coordinates most reliably (but spuriously and weakly), while the learner that *can* react competes by default and colludes only slowly (but genuinely and strongly). This reconciles the apparently opposed findings that “simple learners collude” and that “Q-learning collusion is fragile/slow”—they describe the two faces of the same memory margin. Second, the genuine/spurious distinction is *operational and structural*, not a matter of exploration tuning: a stateless learner cannot punish at any ε , so its coordination is spurious by construction, which the forced-deviation test verifies. Third, the convergence-artifact critique that dominates this literature does not apply to the stateless result, because that learner converges; it applies, correctly, to the stateful learner, whose horizon dependence we therefore trace explicitly.

For enforcement, the reading is detection-centred and, we think, uncomfortable. The coordination that is easiest to produce (it needs no memory, no design, and no patience), arises fastest, and is hardest to distinguish from competition by reaction-based signatures is precisely the *spurious* kind generated by the simplest algorithm. A behavioural test for punishment dynamics would clear the stateless learner while catching the Q-learner, exactly inverting their consumer impact at short horizons. An effects-based screen on supra-competitive *outcomes*—of which our model-free entropy diagnostic is an example—is the appropriate instrument, because it is agnostic to the algorithm

and to whether the coordination is genuine.

8 Limitations and conclusion

The analysis is computational and offers no convergence proofs; multi-agent Q-learning may cycle rather than converge [Waltman and Kaymak, 2008], which our chaotic regime and the falling convergence rates at long horizons reflect. We study homogeneous goods, inelastic unit demand, tabular memory-one learners, and a single market game. Inelastic demand is deliberate—it isolates the coordination mechanism—but it implies zero deadweight loss, so our policy claims concern *detection of supra-competitive pricing*, not welfare magnitudes; an elastic-demand extension is the natural next step, as is replication across Cournot and repeated-game environments to test whether the memory margin is invariant to best-response geometry. Reported regime frequencies depend on the entropy threshold, which we therefore derive from the data and whose sensitivity we report in full.

Subject to these caveats, the picture is clear and honest. In the environment most hostile to collusion, a stateless learner reaches supra-competitive prices immediately, reliably, weakly, and spuriously, while a stateful learner competes by default and learns intense, genuine collusion only slowly. Memory is the margin; reliability and intensity trade off; and because the most reliable coordination is the most spurious and least detectable, the case for outcome-based, algorithm-agnostic screening is strengthened rather than weakened.

Data and Code Availability

All simulation code, configuration files, fixed random seeds, and generated Monte Carlo output are openly available in the project repository at <https://github.com/shubhL-research/algorithmic-bertrand-rl> (archived release with a Zenodo DOI). Every figure and statistic reported here is reproduced from seeds by `src/roadb_experiments.py`.

Declarations

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